



Abstract

Both DC and Marvel have, in recent years, adopted the concept of a “multiverse,” a framework in which an infinite number of parallel universes coexist, each representing a distinct possible outcome of events. Although widely used as a narrative mechanism, the internal structure and theoretical implications of such a model have received limited formal analysis. This paper examines the plausibility of multiversal frameworks as they are presented in contemporary comic and cinematic storytelling, and develops a structured approach for evaluating how branching outcomes might behave within an infinite decision space. We outline several constraints that arise when narrative continuity is situated within a multiversal context and consider how these constraints shape the interpretation of events across parallel timelines.

1. Introduction

The adoption of multiverse models in popular superhero narratives introduces a set of implicit assumptions about causality, contingency, and the completeness of possible outcomes. Under these models, every decision or event generates a divergent path, producing an unbounded collection of parallel timelines. Each timeline represents a distinct and admissible realization of all events up to and beyond the branching point.

While these models have been explored in various narrative forms, their structural consequences for long-form character arcs remain underexamined. This paper proposes a formal approach to interpreting character trajectories within such a multiversal space.

2. Branching Outcome Framework

Let the set of all possible universes be denoted U , where each $u \in U$ corresponds to a complete sequence of events. At each decision point D , a branching surface generates the full set of outcomes $\{d_1, d_2, \dots, d_n\}$, all of which occur with nonzero measure across U .

This assumption produces a decision manifold in which:

1. Every admissible outcome is instantiated.
2. No branch is privileged at the point of divergence.
3. Global outcome completeness is preserved, regardless of local probability weighting.

Within such a space, any sequence of events—no matter how unlikely—necessarily exists in at least one universe. This raises the possibility that apparent narrative exceptionalism may instead reflect a selection effect over survivable trajectories.

3. Observability and Narrative Propagation

A key feature of narrative continuity is that it requires the events of a given timeline to persist long enough for the narrative to be transmitted. Whether through direct diegetic documentation (e.g., eyewitnesses, recordings) or extra-diegetic mechanisms (e.g., the “camera” of cinematic presentation), a timeline becomes part of the observable corpus only if it avoids terminal failure up to the point of transmission.

$$S(u) = \begin{cases} 1 & \text{if } u \text{ is observable} \\ 0 & \text{otherwise} \end{cases}$$

Define:

$$O = \{u \in U \mid S(u) = 1\}$$

Thus, for any timeline u , observability is contingent upon the existence of a sequence of non-terminal states sufficient to allow the record to be generated and received. Let $O \subseteq U$ denote the subset of universes satisfying this observability condition.

By construction, all narratives available to readers or viewers are drawn exclusively from O , regardless of the relative measure of O within the full multiverse.

4. Survivor Trajectory Convergence

Given infinite branching, the set of universes in which any particular sequence of non-terminal outcomes occurs is guaranteed to be non-empty. Even outcomes with extremely low local probability retain representation somewhere within U . Consequently, for any character and any chain of events, there always exists at least one timeline in which that chain proceeds without catastrophic termination.

Narrative presentation therefore exhibits a strong convergence toward these non-terminal trajectories—not because they are typical, but because only these trajectories can enter the observable set.

This produces the appearance of coherent, continuous character arcs even when the underlying probability structure of the multiverse would otherwise predict rapid termination.

5. Conclusion

When narrative continuity is embedded within an infinite multiverse of branching outcomes, the apparent success, survival, or exceptional achievement of a character does not reflect properties intrinsic to the character. Instead, it reflects the observer's sampling from the subset of timelines in which events proceeded without premature collapse.

Heroism, in this context, is not a causal property of the character, but a statistical artifact of observer selection.

Heroism is an expression of survivor bias.

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